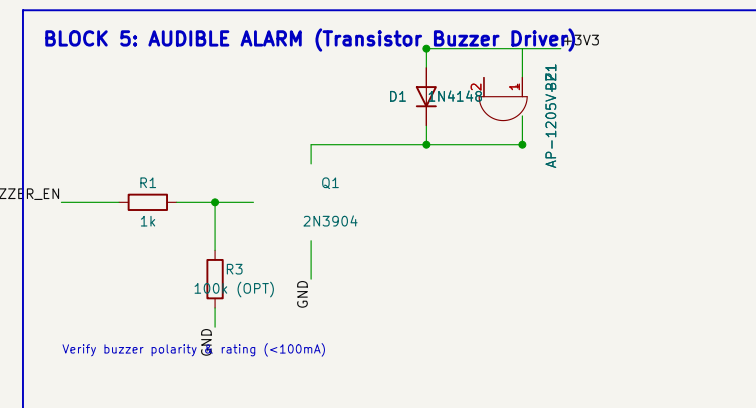
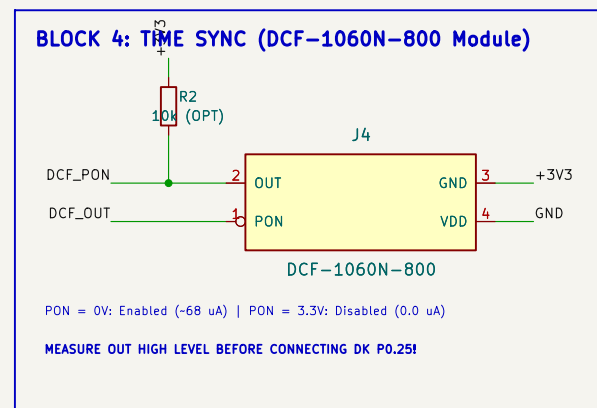
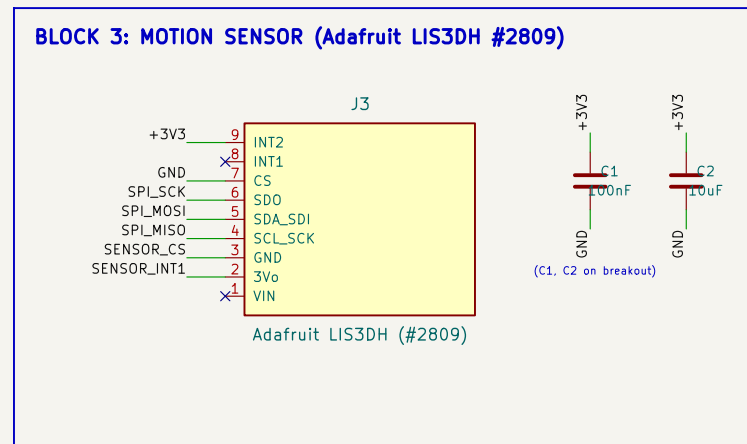
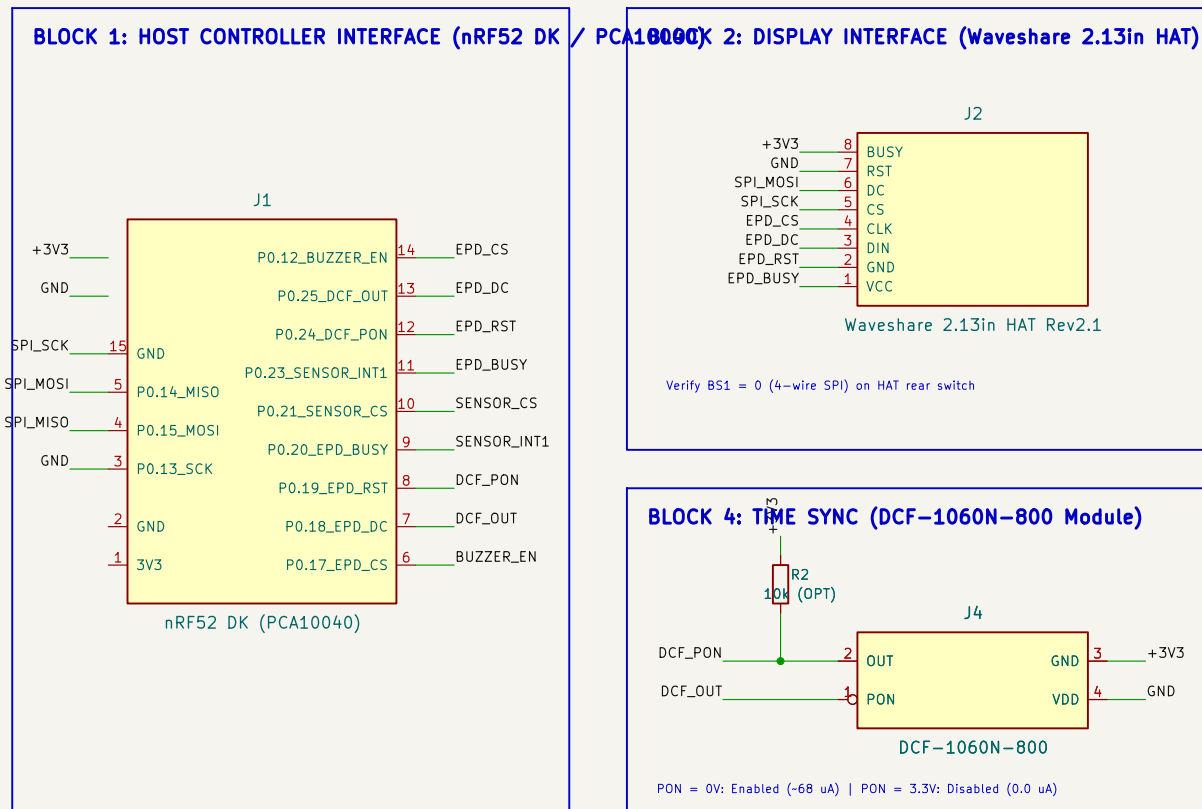




PROTOTYPE SAFETY & INTEGRATION RULES

1. POWER: Power prototype solely from DK 3V3 rail via USB. NEVER connect KORAD bench supply and DK 3V3 rail together!
2. SAFE CONTROL STATES: Set DCF_PON HIGH (disabled, 0.0uA). Set EPD_CS and SENSOR_CS HIGH (unselected). Set BUZZER_EN LOW (off).
3. DCF-77 VERIFICATION: Measure DCF OUT high-level voltage before connecting to DK P0.25. Orient antenna perpendicular to Frankfurt.
4. E-PAPER CONFIGURATION: Verify switch BS1 = 0 on back of Waveshare HAT for 4-wire SPI mode before powering display rail.
5. BUZZER DRIVER: Verify 2N3904 TO-92 pinout (1=Emitter, 2=Base, 3=Collector). Diode D1 cathode connects to +3V3 to clamp flyback spikes.
6. STAGED BRING-UP: 1) DK Blinky -> 2) E-Paper Demo -> 3) LIS3DH WHO_AM_I -> 4) DCF Standalone -> 5) DCF Sync -> 6) Buzzer Alarm.

nRF52 DK (PCA10040) + E-Paper + LIS3DH + DCF-77 + Buzzer Driver
Bachelor Thesis (APD)

Sheet: /
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Title: Ultra-Low-Power Bedside Clock – Breadboard Prototype

Size: A4	Date: 2026-09-13
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